# Pruthvi Prodduturi

Engineering Lead

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## PROFESSIONAL SUMMARY


Senior software engineer with 11+ years across distributed data systems, query engines, and zero-to-one platform engineering. Authored a distributed SQL engine (MPP) performance analysis that was escalated from the Azure Data President to CEO Satya Nadella and influenced the engine roadmap. Built the governed lakehouse behind Microsoft's AI competitive intelligence — billions of rows/month, 50+ stakeholders — and designed a real-time IoT orchestration platform from zero for autonomous datacenter robots, with environmental monitoring over 18 PB/month of datacenter telemetry (~630B rows/day). Fluent from query-engine internals to cloud-native service architecture, and effective communicating technical direction to executive audiences.


## CORE SKILLS

**Distributed Systems:** MPP query engines, columnar storage, vectorized execution, partitioning strategies, async job systems, IoT orchestration

**Data Architecture:** Lakehouse design, semantic models, telemetry pipelines, ML scoring, stream/batch processing

**Data Engineering:** SQL optimization, Presto, Trino, StarRocks, ClickHouse, Fabric SQL, ADF, Synapse, ADLS, Cosmos DB

**Platform Engineering:** Python, FastAPI, React, Azure (App Service, IoT Hub, Blob Storage), managed identity, VNet

**Leadership:** Zero-to-one platform builds, cross-team API contracts, executive technical communication, operating under ambiguity

## EXPERIENCE

### Microsoft *Redmond, WA*

**Senior Software Engineer** Nov 2022 – Present

#### Distributed Systems & Query Engine Analysis

- **Authored** a deep performance analysis of Microsoft's distributed SQL engine (MPP architecture, columnar storage) covering query optimization, predicate pushdown, partition pruning, and materialization strategies. Escalated from the Azure Data President to CEO Satya Nadella. Directly influenced the engine roadmap.
- **Hands-on production experience** across distributed query engines — Fabric SQL, Presto, Trino, StarRocks, ClickHouse, Apache Kylin — with a working understanding of the tradeoffs between MPP, vectorized execution, columnar storage, and lakehouse query patterns from building on all of them.

#### Data Platform Architecture (Billions of Rows / Month)

- **Built** the lakehouse architecture (ingestion → transformation → semantic model → serving) powering Microsoft's AI competitive intelligence — replaced 5+ fragmented org-specific pipelines with a single governed platform serving 50+ stakeholders.
- **Architected** pipelines processing billions of rows/month across 10+ GenAI products (Copilot, ChatGPT, Gemini, Claude, Perplexity), standardizing desktop, mobile, SSO, and connector signals into governed, query-ready models.
- **Established** a unified metrics framework — KPI computation, LLM evaluation workflows, and ML-driven scoring consumed in weekly executive scorecards driving Microsoft's AI competitive strategy.
- **Delivered** sub-5-second query latency on competitive dashboards by designing a semantic model that eliminated scan-heavy patterns — replacing a system of ad-hoc queries and spreadsheets with governed, reproducible analytics.
- **Reduced** SLA breaches by 25% through pipeline reliability engineering and data governance hardening.

### Platform Engineering — IoT Fleet Orchestration (2026)

- **Designed and built** a real-time orchestration platform from zero for autonomous datacenter robots — 26-endpoint REST API (Python/FastAPI), React command center, per-device job queue, and mission lifecycle engine controlling physical hardware via IoT Hub direct methods.
- **Instrumented** real-time environmental monitoring over 18 PB/month of datacenter telemetry (~630 billion rows/day) across distributed datacenter sites.
- **Defined** cross-team API contracts between the cloud platform and robotics firmware team, replacing a CLI-as-API approach with proper service boundaries — working through undocumented firmware behavior, schema changes, and hardware version fragmentation.
- **Engineered** parallel fleet discovery (asyncio + thread pooling), reducing O(n) sequential device calls to a single parallel call and architected to scale to 1000+ devices. Built auth with token deduplication — 8 concurrent requests resolved to one identity call via async futures.
- **Hardened** the production architecture to zero-key operation — managed identity for all service-to-service auth, private endpoints for storage, and VNet integration for network isolation.

**Software Engineer II** Jan 2019 – Nov 2022

- **Delivered** business metrics, data pipelines, and distributed reporting systems for Microsoft News.
- **Built** durable OLAP solutions using Apache Kylin, Mondrian, and ClickHouse for high-volume analytics.
- **Partnered** with COSINE to build a unified hot-data platform using Azure Kubernetes Service and Presto, improving performance and stability.
- **Accelerated** data availability by 7+ hours/day through optimized reporting pipelines.
- **Developed and maintained** CI/CD pipelines, improving deployment consistency and reducing operational overhead.

## EARLIER EXPERIENCE

**Technical Lead** — HCL America 2017 – 2019

Reduced ETL processing time by 95% using C#, Avocado, and SCOPE automation; led offshore telemetry and analytics teams supporting MSN.

**Senior Software Engineer** — Mindtree 2013 – 2016

Delivered BI analytics and telemetry systems for Microsoft IDC; built SCOPE-based ETL pipelines reducing reporting time by 85%; migrated Power Pivot to Power BI, reducing analysis time by 80%.

## EDUCATION

**Kent State University** — M.S., Digital Sciences (Computer & Information Sciences) 2016 – 2017

**Jawaharlal Nehru Technological University** — B.Tech., Electronics & Communication Engineering